

Mitochondrial DNA and Its Applications in Forensic Medicine and Medical Diagnosis: A Comprehensive Analysis

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Authors' contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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Abstract

Mitochondrial DNA (mtDNA) research holds crucial significance for understanding fundamental biological mechanisms and has wide-ranging applications in forensic science and clinical diagnosis. This paper offers an extensive overview of mitochondria, the structure of mitochondrial DNA, the DLoop control region, the 12S gene, mtDNA copy number, heteroplasmy, maternal inheritance, mtDNA replication processes, the genetics of mitochondrial diseases, genome analysis, and NextGeneration Sequencing (NGS) platforms. Furthermore, the review explores the forensic utility of mtDNA, covering kinship analysis, investigative applications, medical diagnostics, mtDNA quantification, differentiation between human and animal blood, Polymerase Chain Reaction (PCR), and mtDNA sequencing. The analysis indicates that mitochondrial DNA serves as a reliable marker for tracing the ancestry of various populations. It is a vital area of research across

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disciplines such as evolutionary anthropology, population genetics, medical genetics, genetic genealogy, and forensic science. The expression of mtDNA is essential for generating ATP via oxidative phosphorylation across all eukaryotic organisms. These mitochondrial structures are thought to link mtDNA replication and transcription mechanisms. Research involving ancient DNA has established a strong correlation between the availability and nutritional composition of diet and diversity in natural populations. In recent years, the emergence of Next-Generation Sequencing (NGS) technologies has instigated a fast and definitive shift, advancing scientific inquiry and contributing substantially to deeper biological understanding. This comprehensive review concludes that mitochondrial DNA is a highly effective and potent Instrument for individual identification, tracing population origins, and diagnosing genetic disorders, thereby pioneering new research and practical applications In these critical areas.

Keywords: Mitochondrial DNA; forensic medicine; medical diagnosis; heteroplasmy; polymerase chain reaction; next-generation sequencing.

1. Introduction

Mitochondria, crucial organelles found within eukaryotic cells, are frequently termed the 'cellular power plants' due to their critical function in generating adenosine triphosphate (ATP) via oxidative phosphorylation (Lavrov and Pett, 2016). Nevertheless, their roles encompass more than just energy generation, extending to the synthesis of amino acids and steroids, oxidation of fatty acids, iron-sulfur metabolism, and triggering apoptosis. The quantity of mitochondrial DNA (mtDNA) copies per mitochondrion and the total number of mitochondria per cell fluctuate depending on the cell type. Typically, each mitochondrion contains 2-10 mtDNA copies, and a single cell can contain hundreds to thousands of mitochondria (Du et al., 2025). The human mitochondrial genome is a circular molecule of approximately 16,569 base pairs, which codes for 13 proteins, 22 tRNAs, and 2 rRNAs, and is inherited exclusively from the mother (Árnadóttir et al., 2024).

Mitochondria are central to the processes of cellular survival and death, possessing their own distinct, small genome separate from the nuclear DNA (Liang et al., 2024). Mitochondrial DNA is highly effective for establishing the ancestry of diverse populations and represents a vital area of investigation In disciplines including evolutionary anthropology, population genetics, medical genetics, genetic genealogy, and forensic science (Pakendorf and Stoneking, 2005; Shriver and Kittles, 2004). Their main role is producing energy as ATP through the citric acid cycle and the oxidative phosphorylation system (OXPHOS). The widespread use of

mitochondrial DNA in phylogeographical research stems from its key characteristics: strictly maternal inheritance, a consistent substitution rate resembling a molecular clock, a high copy number, and an elevated mutation rate coupled with a lack of recombination (Diwedi et al., 2020).

Mitochondria are also involved in the biosynthesis of many metabolites, such as pyrimidines, amino acids, or cellular iron-sulfur cluster proteins (Bereiter-Hahn, 1990; Lill et al., 1999). ATP synthesis occurs through the process of oxidative phosphorylation (OXPHOS) via the respiratory or electron transport chain (ETC), located in the inner mitochondrial membrane and composed of five protein complexes (Zapico and Ubelaker, 2013). Dysfunction in OXPHOS leads to increased production of reactive oxygen species (ROS) (Camello-Almaraz et al., 2006), and mitochondria control the cell's ability to generate and detoxify ROS (Nicholls et al., 2003).

The importance of this study lies in highlighting the increasing role of mitochondrial DNA in understanding diseases and developing diagnostic and criminal investigation techniques. With the rising mortality and losses caused by mitochondrial-related genetic diseases, research in this area becomes crucial for developing more accurate and effective therapeutic and diagnostic solutions. For example, these studies can contribute to identifying individuals at risk of certain diseases or understanding drug resistance mechanisms (Gorman et al., 2015).

nontraditional solutions in identification and medical diagnosis show limitations, especially when dealing with degraded samples or small

amounts of DNA. In contrast, recent studies on mitochondrial DNA offer innovative solutions due to its unique characteristics, such as high copy number and maternal inheritance, making it ideal for analysing challenging samples (Amorim et al., 2019). This comparison highlights the feasibility of adopting modern mitochondrial DNA-based technologies.

Previous studies have addressed various aspects of mitochondrial DNA. For instance, a study by Pakendorf and Stoneking (2005) focused on using mtDNA to determine the origins of populations, while a study by Galtier et al. (2009) investigated the use of mtDNA genes as molecular markers in molecular ecology. These studies pave the way for a deeper understanding and broader applications of mitochondrial DNA.

This current study aims to provide a comprehensive and updated review of the latest developments in understanding mitochondrial DNA and its applications, with a focus on aspects not sufficiently explored in previous studies. It also aims to clarify how technologies like NextGeneration Sequencing (NGS) can revolutionise the fields of forensic medicine and medical diagnosis, providing a reference framework for researchers and practitioners in these areas.

2. Mitochondria

Mitochondria are organelles enclosed by a double membrane and are present in the majority of eukaryotic cells. They are widely known as the cell's 'power plants' due to their function in generating ATP via oxidative phosphorylation, but they also execute several other cellular tasks, including the biosynthesis of amino acids and steroids, β -oxidation of fatty acids, Fe metabolism, and the induction of apoptosis (Lavrov and Pett, 2016). The number of mtDNA copies per mitochondrion and the total mitochondria count per cell is cell-type dependent. Specifically, each mitochondrion typically holds 2–10 mtDNA copies, and a cell can contain between hundreds and thousands of mitochondria (Manuscript, 2014). These organelles are pivotal to the cycle of cellular life and death and possess a small genome, mtDNA, that is distinct from the nuclear DNA. Mitochondrial DNA serves as an effective instrument for ascertaining the geographic origins of various populations. It constitutes a key area of investigation in diverse fields such as evolutionary anthropology, population genetics, medical genetics, genetic genealogy, and forensic science (Pakendorf and Stoneking 2005; Shriver and Kittles, 2004). While their primary role is energy synthesis in the form of ATP through the

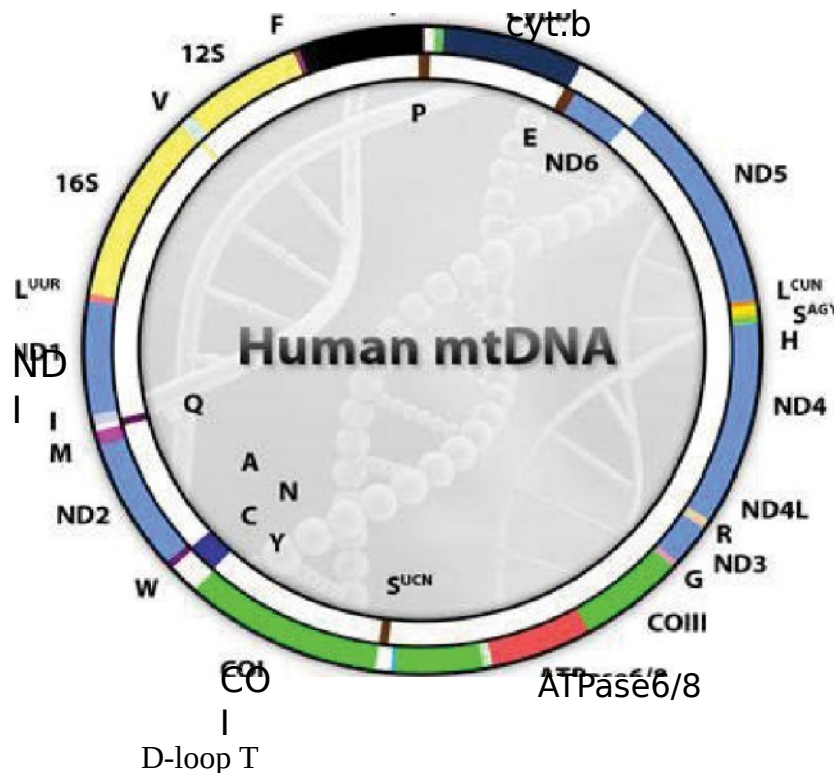
citric acid cycle and the oxidative phosphorylation system (OXPHOS), they additionally participate in the production of various metabolites, such as pyrimidines, amino acids, or cellular iron-sulfur cluster proteins (Bereiter-Hahn, 1990; Lill et al., 1999). ATP is synthesized through oxidative phosphorylation (OXPHOS) within the respiratory or electron transport chain (ETC), which is situated in the inner mitochondrial membrane and comprises five protein complexes (Zapico and Ubelaker, 2013).

3. Mitochondrial DNA Genome

The genetic regulation of mitochondria is overseen by both the mitochondrial and nuclear genomes. The mitochondrial genome is composed of numerous copies of the doublestranded mitochondrial DNA (mtDNA) molecule, which is 16,569 base pairs long and is situated near the OXPHOS system within the matrix (Ng & Turnbull, 2016). A strong collaborative relationship exists between the mitochondrial and nuclear genomes in encoding the subunits for the various electron transport system (ETS) complexes: Complex I (CI) subunits—seven in total (ND1, ND2, ND3, ND4, ND4L, ND5, ND6)—are encoded by the mtDNA; Complex II (CII) is solely encoded by the nuclear DNA; Complex III (CIII) includes cytochrome b (cyt b), a single mitochondrial subunit; Complex IV has three subunits (COI, COII, COIII) encoded by the mtDNA; and Complex V is also partially encoded by the mitochondria with two subunits.

4. Control Region of mtDNA D-Loop

The function of OXPHOS can be compromised by damage to mtDNA or alterations in its copy number. The susceptibility of mtDNA to damage, and consequently to mutations, is heightened by numerous contributing factors. During the replication process, mtDNA features a brief triplestranded DNA formation, termed the D-loop, where one strand of the mtDNA is temporarily displaced (Manuscript, 2014). The control region, also known as the D-loop, is the most variable segment of the mtDNA. As the largest noncoding section of the mtDNA, it is instrumental in governing and initiating both replication and transcription. The control region's most polymorphic sequences are Hypervariable Segment I (HVS I) and Hypervariable Segment II (HVS II). These segments are frequently studied in research concerning population origins and human evolutionary history (Tully et al., 2000; Horai et al., 1995).



mammals consist of the mtDNA-encoded 12S and 16S rRNAs, alongside over one hundred distinct ribosomal proteins encoded by the nuclear genome. In bacterial systems, modifications to rRNA are implemented at specific stages of ribosomal assembly and contribute to rRNA processing, folding, and interactions with adjacent proteins.

6.

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Fig. 1. Organisation the human mitochondrial genome

5. Gene 12S and its uses

For ATP generation via oxidative phosphorylation to occur in all eukaryotes, mtDNA expression is indispensable. Mammalian mtDNA codes for 13 polypeptides, which are synthesized by mitochondrial ribosomes (MIT ribosomes) and are vital for the oxidative phosphorylation process. The MIT ribosomes in

Mitochondrial DNA Copy Number

The estimated count of mtDNA copies per mitochondrion ranges from two to ten. The mtDNA is condensed into nucleoids, which are protein-DNA complexes. Each mitochondrion can house anywhere from one to over ten nucleoids. It is hypothesized that nucleoids reside in specific membrane-spanning entities

called mitochondrial replisomes. These formations are believed to offer a means of coupling mtDNA replication with transcription. While the exact makeup of nucleoids remains largely unclear, a significant number of nuclear-encoded proteins have been identified that regulate mtDNA replication, mtDNA transcription, mitochondrial fusion, and the anchoring of nucleoids to the cytoskeleton (Baron, 2010).

7. Heteroplasmy of MtDNA

Mitochondrial heterogeneity is defined as variations in a mitochondrial characteristic both within the mitochondrial population of a single cell and, potentially when aggregated at the cellular level, among cells that are otherwise considered identical. This heterogeneity has been identified as a significant factor correlating with extrinsic cellular noise. Research revealed that mitochondrial mass, when normalized to the mitochondrial membrane potential, shows a strong correlation with the global transcription rate at the single-cell level, and this factor accounts for approximately 50% of the variability observed in protein levels. Given that the global transcription rate influences various aspects of cellular function, this connection provides strong support for the critical role of mitochondrial heterogeneity in contributing to cellular noise (Aryaman et al., 2019).

8. Mitochondrial Maternity

Within a spermatozoon, mitochondria are situated in the midpiece. During fertilization, only the sperm head, which carries the nucleus but lacks mitochondria, penetrates the egg. The resulting zygote thus inherits the maternal mitochondria, which are then passed on to the progeny. Although paternal mitochondria might occasionally enter the cell, those from the spermatozoon are typically degraded by the egg cell following fertilization. Consequently, the simultaneous inheritance of both maternal and paternal mitochondrial DNA within one person is exceptionally uncommon in humans. A valuable characteristic for forensic identification is that the mtDNA sequence remains identical for individuals who share the same maternal line, allowing for their identification by comparison with maternal relatives (Richard et al., 2015).

9. Replication of Mitochondrial DNA Molecules

At the onset of replication at the initial origin, both strands of mtDNA separate. DNA synthesis then proceeds in one direction along a single

strand, while the opposing single strand is kept excluded until the second replication origin becomes accessible. Subsequently, replication commences in the reverse direction (Krishnan et al., 2008; Shadel and Clayton, 1997). An alternative strand-coupled mechanism is described by the bidirectional replication model. In this model, replication begins at a single replication zone and advances symmetrically in both directions (Bowmaker et al., 2003; Holt et al., 2000; Krishnan et al., 2008; Yao Yang et al., 2002).

10. Genetics of Mitochondrial Diseases

Ascertaining the exact prevalence of mitochondrial disease is challenging, primarily due to the wide clinical and genetic diversity observed. The range of phenotypes associated with mitochondrial disease has broadened considerably since the initial descriptions linking classical clinical syndromes to point mutations and rearrangements in mitochondrial DNA. Advances in genetic technologies have enabled a detailed analysis of the nuclear genome, which has significantly enhanced the diagnostic process for mitochondrial disorders (Gorman et al., 2015). These diseases are characterized by impaired function of the respiratory chain and typically manifest as multi-system and multiorgan conditions, with the muscles and nerves being predominantly affected.

11. Mitochondrial Genome Analysis

Over the course of evolution, the human genome, including its mitochondrial component, has developed genetic adaptations related to dietary intake; these adaptations are instrumental for mapping evolutionary trajectories across space and time. Investigations into ancient DNA have established a robust connection between the dietary availability, nutritional composition, and diversity within natural populations (Perry et al., 2007; Sjöstrand et al., 2014; Tishkoff et al., 2007; Schlebusch et al., 2013). Furthermore, the metabolic function of mitochondria offers another crucial perspective for the analysis of ancient mitochondrial DNA to determine the historical origins of a population (Nesheva, 2014).

12. Next-generation Sequencing Technology (NGS)

The introduction of Next-Generation Sequencing (NGS) technologies in recent years has instigated a fast and definitive shift, addressing the evolving requirements of the scientific

community and profoundly advancing the comprehension of biological processes. These innovative methodologies have transformed how researchers acquire genetic data from biological specimens, fostering new research areas dedicated to the complete characterization of the biological subject under investigation (Cocozza et al., 2014).

13. Mitochondrial DNA in Forensics

Mitochondrial DNA possesses several properties that make it highly valuable for forensic analysis, particularly its lack of recombination, high copy number, and matrilineal inheritance pattern. mtDNA profiling, which relies on sequencing the control region or performing full genomic sequence analysis, is employed for the examination of challenging forensic samples, including ancient bones, teeth, and hair. In subtraction to other biological samples with limited DNA content. The careful interpretation and presentation of results necessitate thorough consideration of biological factors, as well as matters concerning nomenclature and the use of reference population databases (Amorim et al., 2019).

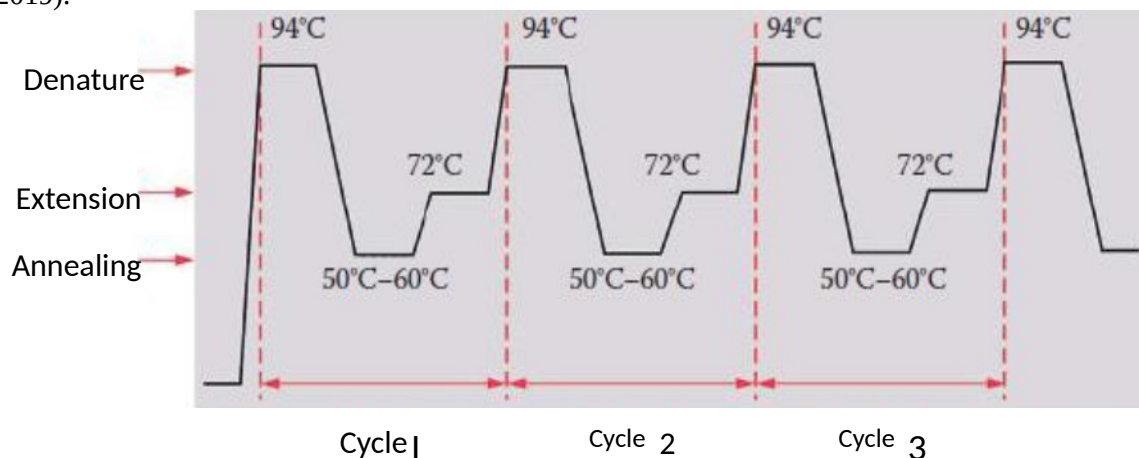


Fig. 2. Temperature parameters during thermal cycling of the PCR process (Richard et al., 2015)

14. Differentiation between Human Blood and Animal Blood

The original researchers propose that utilizing a specific sequence could precisely solve the problem of species differentiation, though they caution that this must be executed with care to prevent overfitting. Subsequently, they conducted external validation using bovine (cow) samples and ten blindly selected unknown samples (comprising 5 human and 5 animal samples), achieving 100% accuracy, even for a small number of samples that were borderline.

These same authors later broadened their dataset to incorporate blood samples from six additional animal species: chimpanzee, deer, elk, ferret, fish, and macaque.

15. Polymerase Chain Reaction (PCR)

Polymerase Chain Reaction (PCR) stands out among the DNA-based methods utilized for species identification. This technique allows for the detection of minute quantities of nucleic acid probes and the determination of their sequence by amplifying DNA or RNA from specific strains. PCR offers several advantages, such as exceptional sensitivity and rapid execution even with a large throughput of samples (Rodriguez et al., 2004).

16. Sequencing Mitochondrial DNA

The process of sequencing a particular mitochondrial DNA region involves employing a combination of PCR amplification and DNA sequencing methods. This integrated approach aims to minimize the time and manual effort required to acquire DNA sequences from genomic DNA templates. Typically, mtDNA

sequencing involves several steps: PCR amplification, followed by DNA sequencing reactions, separation via electrophoresis, and finally, data collection and sequence analysis (Richard et al., 2015).

17. Forensic Uses of Mitochondrial DNA

17.1 Kinship Matching

Mitochondrial DNA profiling is routinely applied in scenarios involving mass disasters, missing persons investigations, and the analysis of historical remains. Such cases often involve

degraded or aged samples, and mtDNA provides a means to determine an Individual's identity by tracing their maternal lineage. Therefore, identifying individuals IS commonly supported by cross-referencing samples from their maternal relatives (Wai, 2024).

17.2 Investigative Use

Direct comparison of an mtDNA sequence with a reference sample can help identify an individual, but this method may be insufficient when only an unknown mtDNA profile is available. In such instances, obtaining information about the unknown source's ancestry and ethnicity is highly beneficial. Inferring maternal Biogeographical Ancestry (BGA) from mtDNA offers valuable investigative leads regarding potential suspect identities. Similar to forensic DNA phenotyping (FDP) techniques, BGA is particularly useful In situations where crime scene samples do not match any known person in a database, and in the review of 'cold cases' (Ellison et al., 2006).

17.3 Medical Diagnostics

Mitochondria are organelles situated within eukaryotic organisms. Their functions at the cellular level encompass energy generation, the metabolism of amino acids, fatty acids, and cofactors, as well as cell signalling pathways. Variations (polymorphisms) within the mitochondrial genome have been associated with differences In endurance capacity and trainability, in addition to various mitochondrial diseases. The clinical diagnosis of these diseases integrates maternal genetic history, clinical signs, biochemical testing, and specific targeted sequencing assays (Wai, 2024).

17.4 Quantification of Mitochondrial DNA

MtDNA quantification is performed to establish the amount of 'amplifiable' DNA present in a biological extract. This measurement is crucial for optimizing the DNA input for amplification, thereby ensuring the process's overall success. Insufficient DNA amplification risks causing allele dropouts, heterozygote imbalance, and inadequate sequence coverage. Conversely, excessive DNA amplification can result in the appearance of stutter peaks, non-specific amplification, and a reduction in the consistent representation of coverage across different samples (Robin et al., 2016).

18. Conclusions

In summary, this comprehensive review establishes mitochondrial DNA as an exceptionally valuable genetic asset for forensic medicine and clinical diagnosis. It offers distinctive perspectives on population origins, personal identification, and the mechanisms of disease, thus paving the way for exciting new avenues in scientific research and development.

Consent

It is not applicable.

Ethical Approval

It is not applicable.

Disclaimer (Artificial Intelligence)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

Competing Interests

Authors have declared that no competing interests exist.

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